

$$r = 3$$

We know that for median of medians once selected half of the arrays will have at least 2 elements smaller than the median:

$$2 \cdot \left(\frac{1}{2} \cdot \frac{n}{3}\right) = \frac{n}{3}$$

The worst case is then if the actual median is in the other portion

$$n - \frac{n}{3} = \frac{2n}{3}$$

And the problem is divided into sub arrays of 3 elements is

$$\frac{n}{3}$$

$$T(n) \leq T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + O(n)$$